AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

Chemistry Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are SEVEN questions. Answer ALL SEVEN questions. Questions 6 & 7 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 5 Marks)

a. The following reactions occurred during the industrial preparation of hydrogen.

Equation A: $C_{(s)} + H_2O_{(g)} \Leftrightarrow [X]_{(g)}$

Equation B: $[X]_{(g)} + H_2O_{(g)} \xleftrightarrow[1000^{\circ}C]{\text{Catalyst}} CO_{2(g)} + 2H_{2(g)}$

i. Identify the product "**X**" in equation A. (1 Mark)

_____

ii. Name the catalyst used in equation B. (1 Mark)

_____

iii. How will you separate out hydrogen from carbon dioxide? (1 Mark)

_____

b. A group of people live near a lake, which is the only source of water for that area. Since last year nearby factories have been adding their waste chemicals to it. The people have been drinking the same water and hence suffering from diseases.

i. Name ONE common disease caused by drinking polluted water. (1 Mark)

_____

ii. Suggest ONE method for the purification of polluted water in order to make it drinkable. (1 Mark)

_____

_____

Q.2. (Total 5 Marks)

a. Give reasons why (3 Marks)

i. SiO_2 is used in making laboratory apparatus.

_____

_____

ii. Sodium silicate is used for fire proofing wood.

_____

_____

iii. Silicones are used in plastic surgery and cosmetics.

_____

_____

b. Recognize the oxidizing and the reducing agent from the equation given below. (2 Marks)

$$2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$$

_____

_____

Q.3. (Total 5 Marks)

a. Write the chemical reactions with the help of balanced equations to show the stages of the Contact process.

i. Oxidation of SO_2 gas to SO_3 gas. (3 Marks)

State conditions and type for the given reaction.

_____

_____

ii Production of Oleum. (1 Mark)

_____

_____

b. Chlorine gives chlorides when it reacts with metals. (1 Mark)

Give ONE example for the above reaction.

_____

_____

Q.4. (Total 4 Marks)

Pure iron is obtained by the reduction of ore in the blast furnace.

a. Write a balanced chemical equation to show the reduction of iron. (2 Marks)

_____

_____

b. Why is limestone added into a blast furnace? (1 Mark)

_____

_____

c. Give ONE commercial use of slag. (1 Mark)

_____

_____

Q.5. (Total 6 Marks)

a. Chlorine reacts with methane to give two useful products chloromethane and dichloromethane.

i. Predict a condition necessary for the above reaction. (1 Mark)

_____

_____

ii. Draw the structures for the following: (2 Marks)

Chloromethane	Dichloromethane

b. A person wants to paint his house in order to make his house look beautiful.
Suggest him THREE qualities of a good paint. (3 Marks)

_____

_____

_____

_____

_____

_____

Q.6. (Total 7 Marks)

EITHER

a. Describe the industrial preparation of nitrogen and oxygen. Also give ONE use of each.

OR

b. Discuss the properties of diamond and graphite with reference to their structures. Also give TWO uses of each.

Q.7. (Total 8 Marks)

EITHER

a. Discuss FOUR physical and chemical properties of metals and non-metals.

OR

b. Write FOUR characteristics of alloys which are much recommended. Also name THREE alloys of metal and give uses of each.

END OF PAPER

Please use this page for rough work